STAT A411/611 Lecture Notes

Instructor: Rieken Venema **Office:** SSB 158E
Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45

Today: Sections 5.1 and 5.2.

Monday: Sections 4.4 and 4.5.

Sections 4.1–4.3

For a single random variable X:

- 4.1: X, its values, pdf and cdf, and $U(a, b)$.
- 4.2: Expected values and related quantities:

$$E[h(X)] = \int_{-\infty}^{\infty} h(x)p(x)\,dx, \qquad \mu = E[X], \quad \mathrm{sd}(X).$$

- 4.3: Gamma function and distributions:

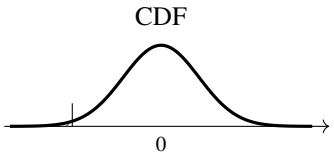
$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1}e^{-x}\,dx, \qquad X \sim \Gamma(\alpha, \beta), \quad X \sim \mathrm{E}(\beta), \quad X \sim \chi^2_\nu \text{ (chi-squared)}.$$

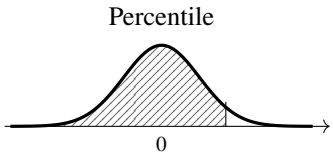
Section 4.4: The Normal Distribution

Normal distribution: $X \sim N(\mu, \sigma^2)$, with a bell-shaped pdf centered at μ. Standardization gives

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

Table V gives the standard normal cdf. Percentiles correspond to cumulative probabilities to the left of a cutoff.





Section 4.5: Chebyshev's Rule and the Empirical Rule

For k standard deviations about the mean, Chebyshev's rule applies to any distribution; the empirical rule applies to approximately normal distributions.

	$k = 1$	$k = 2$	$k = 3$
Chebyshev	—	$\geq 75\%$	$\gtrsim 89\%$
Empirical rule	$\approx 68\%$	$\approx 95\%$	$\approx 99.7\%$

Chapter 5: Joint Probability Distributions

From Univariate to Multivariate Statistics

In Chapters 3 and 4, we studied the probability distribution of a single random variable X (univariate statistics). In practice, statisticians study several interacting random variables and their joint probability distribution: $(X_1, X_2, \ldots, X_k)$.

For the bivariate case ($k = 2$), let X and Y be two random variables and consider the joint distribution of (X, Y).

- Case I: X and Y are both discrete random variables.
- Case II: X and Y are both continuous random variables.

Section 5.1: Joint Distributions

Case I: Two Discrete Random Variables

The joint probability mass function (pmf) $p(x, y)$ of (X, Y) is

$$p(x, y) = P(X = x \cap Y = y).$$

It maps $\mathbb{R}^2$ to $[0, 1]$ and is a valid joint pmf if

$$p(x, y) \geq 0 \quad \text{for every } (x, y), \qquad \sum_x \sum_y p(x, y) = 1.$$

Example 5.1.1: Defective Welds and Improperly Tightened Bolts

Randomly select a car from the production line. Define

$$X = \text{number of defective welds}, \qquad x \in \{0, 1, 2\},$$
$$Y = \text{number of improperly tightened bolts}, \quad y \in \{0, 1, 2, 3\}.$$

Their joint probability distribution is given in Table 5.1.

	$Y = 0$	$Y = 1$	$Y = 2$	$Y = 3$	$p_X(x)$
$X = 0$	0.84	0.03	0.02	0.01	$p_X(0)$
$X = 1$	0.06	0.01	0.008	0.002	$p_X(1)$
$X = 2$	0.01	0.005	0.004	0.001	$p_X(2)$
$p_Y(y)$	$p_Y(0)$	$p_Y(1)$	$p_Y(2)$	$p_Y(3)$	

$$p(0, 0) = P(X = 0 \cap Y = 0) = 0.84.$$

For example,

$$p(1, 2) = P(X = 1 \cap Y = 2) = 0.008,$$
$$P(X = 1) = 0.06 + 0.01 + 0.008 + 0.002 = 0.08.$$

The joint probabilities sum to 1.

Marginal Probability Distributions: Discrete Case

The distribution of X *alone*, obtained by summing over the possible values of Y, is the marginal pmf of X. Similarly for Y:

$$p_X(x) = P(X = x) = \sum_y p(x, y), \qquad p_Y(y) = P(Y = y) = \sum_x p(x, y).$$

These correspond to the row and column totals of Table 5.1.

Case II: Two Continuous Random Variables

The joint probability distribution of continuous random variables X and Y is given by their joint probability density function $p(x, y)$. It satisfies

$$p(x, y) \geq 0, \qquad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p(x, y)\, dx\, dy = 1.$$

For a rectangular region in the (x, y) plane,

$$P(a \leq X \leq b \ \cap \ c \leq Y \leq d) = \int_c^d \int_a^b p(x, y)\, dx\, dy.$$

Marginal Probability Densities: Continuous Case

Integrating the joint pdf over the other variable yields the marginal pdfs:

$$p_X(x) = \int_{-\infty}^{\infty} p(x,y)\,dy, \qquad p_Y(y) = \int_{-\infty}^{\infty} p(x,y)\,dx.$$

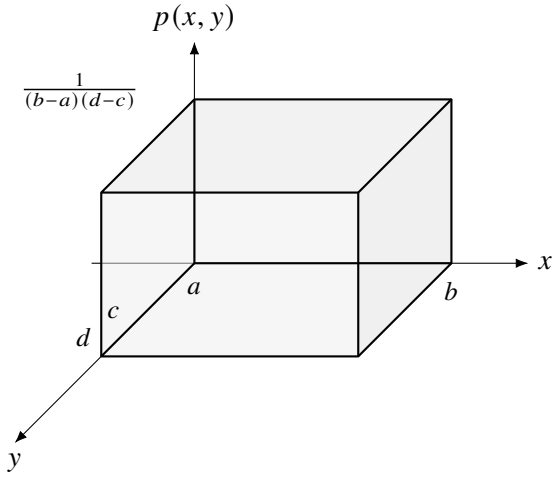
Bivariate Continuous Uniform Distribution

Definition. (X,Y) has a uniform distribution with parameters a, b, c, d, written

$$(X,Y) \sim U(a,b,c,d),$$

if its joint pdf is constant on the rectangle:

$$p(x,y) = \begin{cases} \dfrac{1}{(b-a)(d-c)}, & a \le x \le b,\ c \le y \le d, \\ 0, & \text{elsewhere.} \end{cases}$$



Example 5.1.3: Calcium and Cholesterol Levels

For the continuous uniform distribution

$$(X, Y) \sim U(8.5, 10.5, 120, 240),$$

let X denote the calcium level in the blood and Y denote the cholesterol level in the blood. Their joint pdf is

$$p(x, y) = \begin{cases} \dfrac{1}{(10.5 - 8.5)(240 - 120)} = \dfrac{1}{240}, & 8.5 \le x \le 10.5,\ 120 \le y \le 240, \\ 0, & \text{elsewhere.} \end{cases}$$

The marginal distributions are

$$X \sim U(8.5, 10.5), \qquad Y \sim U(120, 240).$$

Definition: Independence

Two random variables X and Y are called independent if their joint probability function factors into the product of the two marginal functions:

$$p(x, y) = p_X(x)\, p_Y(y) \quad \text{for all } (x, y).$$

The definition applies to discrete *pmfs* and to continuous *pdfs*.

Example 5.1.1 is not independent:

$$p(0, 0) = 0.84 \neq (0.90)(0.91) = p_X(0)p_Y(0).$$

Thus X and Y are dependent in the production-line example.

Example 5.1.3 is independent: within the uniform support,

$$p(x, y) = \frac{1}{240} = \frac{1}{2} \cdot \frac{1}{120} = p_X(x)p_Y(y).$$

Hence X and Y are independent under this uniform model.

Section 5.2: Expected Values of Functions of Two Random Variables

Comparison with Chapters 3 and 4

Chapter	Basic distribution	Expected value
3	Discrete X	$E[h(X)]$
4	Continuous X	$E[h(X)]$
5	Pair (X, Y)	$E[h(X, Y)]$

Let $h(x, y)$ be any function of two random variables X, Y.

1. Both discrete:
$$E[h(X,Y)] = \sum_x \sum_y h(x,y)p(x,y).$$

2. Both continuous:
$$E[h(X,Y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(x,y)p(x,y)\,dx\,dy.$$

Linearity of Expected Value

Property. For constants a, b, c,

$$E[aX + bY + c] = aE[X] + bE[Y] + c.$$

$E(\cdot)$ is a linear operator. For continuous random variables,

$$\begin{aligned} E[aX + bY + c] &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (ax + by + c)p(x,y)\,dx\,dy \\ &= a\iint xp(x,y)\,dx\,dy + b\iint yp(x,y)\,dx\,dy + c\iint p(x,y)\,dx\,dy \\ &= aE[X] + bE[Y] + c \cdot 1 \\ &= aE[X] + bE[Y] + c. \qquad \square \end{aligned}$$

Definition: Covariance

For two random variables X and Y, the covariance is

$$\mathrm{Cov}(X,Y) = E\big[(X - E[X])(Y - E[Y])\big].$$

Computational formula:

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y].$$

Derivation. For continuous random variables, set $\mu_X = E[X]$ and $\mu_Y = E[Y]$. Expanding,

$$\begin{aligned}
\mathrm{Cov}(X,Y) &= \iint (x-\mu_X)(y-\mu_Y)p(x,y)\,dx\,dy \\
&= \iint xy\,p(x,y)\,dx\,dy - \mu_Y \iint x\,p(x,y)\,dx\,dy \\
&\quad - \mu_X \iint y\,p(x,y)\,dx\,dy + \mu_X\mu_Y \iint p(x,y)\,dx\,dy \\
&= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y \\
&= E[XY] - E[X]E[Y]. \qquad \square
\end{aligned}$$

Interpreting Covariance and Changes of Units

Changing the measurement units of X and Y changes the numerical covariance:

$$\underset{\text{Cov=12.51}}{\{(x_i, y_i)\}_{\circ\mathrm{C}}} \xrightarrow{\text{change of units}} \underset{\text{Cov=0.36}}{\{(x_i', y_i')\}_{\circ\mathrm{F}}}.$$

Proposition: Independence Implies Zero Covariance

If X and Y are independent, then

$$\mathrm{Cov}(X,Y) = 0,$$

and X, Y are called uncorrelated.

For continuous independent random variables, $p(x,y) = p_X(x)p_Y(y)$:

$$\begin{aligned}
E[XY] &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\,p(x,y)\,dx\,dy \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\,p_X(x)p_Y(y)\,dx\,dy \\
&= \int_{-\infty}^{\infty} y p_Y(y)\left(\int_{-\infty}^{\infty} x p_X(x)\,dx\right) dy \\
&= E[X]\int_{-\infty}^{\infty} y p_Y(y)\,dy \\
&= E[X]E[Y].
\end{aligned}$$

Thus, using the computational formula,

$$\begin{aligned}
\mathrm{Cov}(X,Y) &= E[XY] - E[X]E[Y] \\
&= E[X]E[Y] - E[X]E[Y] \\
&= 0. \qquad \square
\end{aligned}$$